Problem

Find all values of a, b and c (or conditions on a, b and c) so that the system

$$\begin{array}{rcrcrcl} 2x & + & 3y & + & az & = & b \\ & - & y & + & 2z & = & c \\ x & + & 3y & - & 2z & = & 1 \end{array}$$

has (i) a unique solution, (ii) no solutions, and (iii) infinitely many solutions. In (i) and (iii), find the solution(s).

$$\left[\begin{array}{ccc|c} 2 & 3 & a & b \\ 0 & -1 & 2 & c \\ 1 & 3 & -2 & 1 \end{array}\right] \xrightarrow{r_1 \leftrightarrow r_3} \left[\begin{array}{ccc|c} 1 & 3 & -2 & 1 \\ 0 & -1 & 2 & c \\ 2 & 3 & a & b \end{array}\right] \xrightarrow{-2r_1 + r_3} \left[\begin{array}{ccc|c} 1 & 3 & -2 & 1 \\ 0 & -1 & 2 & c \\ 0 & -3 & a+4 & b-2 \end{array}\right]$$

$$\xrightarrow{-3r_2 + r_3} \left[\begin{array}{ccc|c} 1 & 3 & -2 & 1 \\ 0 & -1 & 2 & c \\ 0 & 0 & a-2 & b-2-3c \end{array}\right]$$

Case (I): $a-2 \neq 0$ or $a \neq 2$

$$\xrightarrow{\left(\frac{1}{a-2}\right) r_3} \left[\begin{array}{ccc|c} 1 & 3 & -2 & 1 \\ 0 & -1 & 2 & c \\ 0 & 0 & 1 & \frac{b-2-3c}{a-2} \end{array}\right] \xrightarrow[r_2 + 2r_3]{-r_2} \left[\begin{array}{ccc|c} 1 & 3 & -2 & 1 \\ 0 & 1 & 0 & -c+2\left(\frac{b-2-3c}{a-2}\right) \\ 0 & 0 & 1 & \frac{b-2-3c}{a-2} \end{array}\right]$$

$$\xrightarrow{r_1+2r_3}\left[\begin{array}{ccc|c} 1 & 3 & 0 & 1+2\left(\frac{b-2-3c}{a-2}\right) \\ 0 & 1 & 0 & -c+2\left(\frac{b-2-3c}{a-2}\right) \\ 0 & 0 & 1 & \frac{b-2-3c}{a-2} \end{array}\right] \xrightarrow{-3r_2+r_1} \left[\begin{array}{ccc|c} 1 & 0 & 0 & 1+3c-4\left(\frac{b-2-3c}{a-2}\right) \\ 0 & 1 & 0 & -c+2\left(\frac{b-2-3c}{a-2}\right) \\ 0 & 0 & 1 & \frac{b-2-3c}{a-2} \end{array}\right]$$

Case(I): Hence when $a \neq 2$, the solution is $x = 1+3c-4\left(\frac{b-2-3c}{a-2}\right)$, $y = -c+2\left(\frac{b-2-3c}{a-2}\right)$ and $z = \frac{b-2-3c}{a-2}$.

Case (II): when $a-2=0$ and $b-2-3c \neq 0$, then the system has no solution.

$$\left[\begin{array}{ccc|c} 1 & 3 & -2 & 1 \\ 0 & -1 & 2 & c \\ 0 & 0 & 0 & b-2-3c \end{array}\right]$$

Case(III): when $a-2=0$ and $b-3c=2$, the system has infinitely many solutions.

$x = 1-3(-c+2s)+2s = 1+3c-4s$

$y = -c+2s$

(free variable) $z = s$

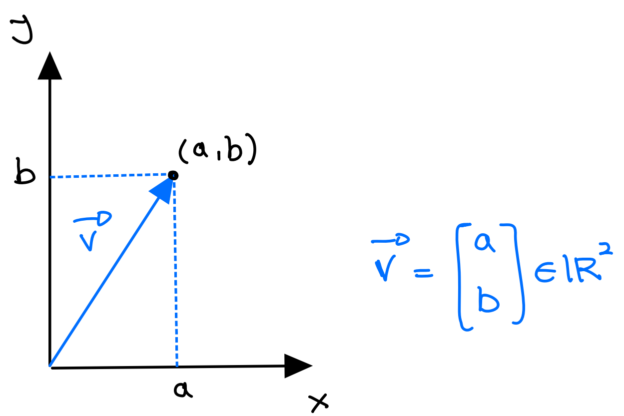
where $s \in \mathbb{R}$.

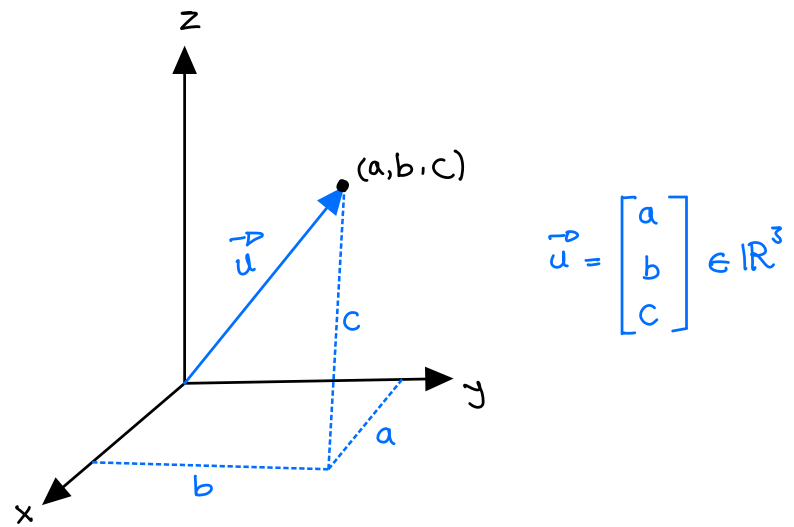
Notation and Terminology

- A column vector is a matrix with only one column.
- $\mathbb{R}$ denotes the set of real numbers.
- $\mathbb{R}^2$ denotes the set of all column vectors with two entries.
- $\mathbb{R}^3$ denotes the set of all column vectors with three entries.
- In general, $\mathbb{R}^n$ denotes the set of all column vectors with n entries.

$\mathbb{R}^2$ and $\mathbb{R}^3$

Vectors in $\mathbb{R}^2$ and $\mathbb{R}^3$ have convenient geometric representations as **position vectors** of points in the 2-dimensional plane and in 3-dimensional space, respectively.





Algebra in $\mathbb{R}^n$

Addition in $\mathbb{R}^n$

- If $\vec{u}$ and $\vec{v}$ are in $\mathbb{R}^n$, then $\vec{u}+\vec{v}$ is obtained by adding together corresponding entries of the vectors.
- The zero vector in $\mathbb{R}^n$ is the $n \times 1$ zero matrix, and is denoted $\vec{0}$.

Example

Let $\vec{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$. Then, $\vec{u}+\vec{v} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} = \begin{bmatrix} 5 \\ 7 \\ 9 \end{bmatrix}$

Properties of Vector Addition

Let $\vec{u}$, $\vec{v}$, and $\vec{w}$ be vectors in $\mathbb{R}^n$. Then the following properties hold.

1. $\vec{u}+\vec{v} = \vec{v}+\vec{u}$ (vector addition is commutative).
2. $(\vec{u}+\vec{v})+\vec{w} = \vec{u}+(\vec{v}+\vec{w})$ (vector addition is associative).
3. $\vec{u}+\vec{0} = \vec{u}$ (existence of an additive identity).
4. $\vec{u}+(-\vec{u}) = \vec{0}$ (existence of an additive inverse).

Scalar Multiplication

If $\vec{u}$ is a vector in $\mathbb{R}^n$ and $k \in \mathbb{R}$ is a scalar, then $k\vec{u}$ is obtained by multiplying every entry of $\vec{u}$ by k.

Properties of scalar Multiplication

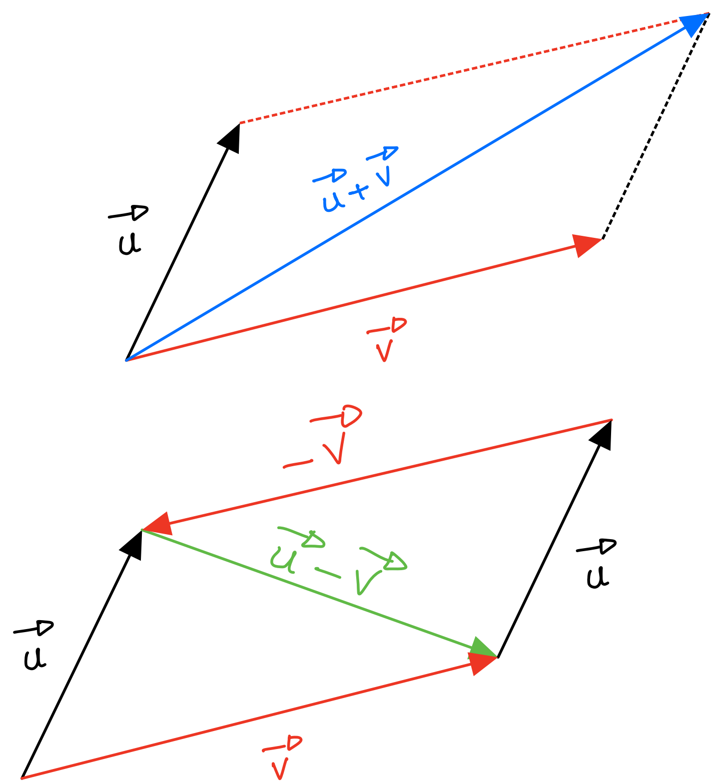
Let $\vec{u}, \vec{v} \in \mathbb{R}^n$ be vectors and $k, p \in \mathbb{R}$ be scalars. Then the following properties hold.

1. $k(\vec{u} + \vec{v}) = k\vec{u} + k\vec{v}$ (scalar multiplication distributes over vector addition).
2. $(k + p)\vec{u} = k\vec{u} + p\vec{u}$ (addition distributes over scalar multiplication).
3. $k(p\vec{u}) = (kp)\vec{u}$ (scalar multiplication is associative).
4. $1\vec{u} = \vec{u}$ (existence of a multiplicative identity).

Example: Let $\vec{u} = \begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$ and $k = -2$. Then $k\vec{u} = -2\begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \\ -10 \end{bmatrix}$.

The Geometry of Vector Addition

Let $\vec{u}$, $\vec{v}$ be vectors. Then $\vec{u}+\vec{v}$ is the diagonal of the parallelogram defined by $\vec{u}$ and $\vec{v}$, and having the same tail as $\vec{u}$ and $\vec{v}$.



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Homogeneous Systems of Equations

Definition

A homogeneous system has the form

$$
\begin{aligned}
a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= 0 \\
a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= 0 \\
&\vdots \\
a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= 0
\end{aligned}
$$

where a_{ij} are scalars and x_i are variables, $1 \leq i \leq m$, $1 \leq j \leq n$.

Notice that $x_1 = 0, x_2 = 0, \cdots, x_n = 0$ is always a solution to a homogeneous system of equations. We call this the trivial solution. We are interested in finding, if possible, nontrivial solutions.

Homogeneous Equations

Example

Solve the system
$$\begin{array}{rcrcrcrcl} x_1 & + & x_2 & - & x_3 & + & 3x_4 & = & 0 \\ -x_1 & + & 4x_2 & + & 5x_3 & - & 2x_4 & = & 0 \\ x_1 & + & 6x_2 & + & 3x_3 & + & 4x_4 & = & 0 \end{array}.$$

$$\left[\begin{array}{cccc|c} 1 & 1 & -1 & 3 & 0 \\ -1 & 4 & 5 & -2 & 0 \\ 1 & 6 & 3 & 4 & 0 \end{array}\right] \xrightarrow[-r_1+r_3]{r_1+r_2} \left[\begin{array}{cccc|c} 1 & 1 & -1 & 3 & 0 \\ 0 & 5 & 4 & 1 & 0 \\ 0 & 5 & 4 & 1 & 0 \end{array}\right] \xrightarrow{-r_2+r_3} \left[\begin{array}{cccc|c} 1 & 1 & -1 & 3 & 0 \\ 0 & 5 & 4 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{\frac{1}{5}r_2} \left[\begin{array}{cccc|c} 1 & 1 & -1 & 3 & 0 \\ 0 & 1 & \frac{4}{5} & \frac{1}{5} & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right] \xrightarrow{-r_2+r_1} \left[\begin{array}{cccc|c} 1 & 0 & -\frac{9}{5} & \frac{14}{5} & 0 \\ 0 & 1 & \frac{4}{5} & \frac{1}{5} & 0 \\ 0 & 0 & \underbrace{0 \quad\; 0}_{} & & 0 \end{array}\right] \longrightarrow$$

$$x_1 - \frac{9}{5}x_3 + \frac{14}{5}x_4 = 0$$
$$x_2 + \frac{4}{5}x_3 + \frac{x_4}{5} = 0$$

x_3 and x_4 are free variables $\longrightarrow$

$$x_3 = s$$
$$x_4 = t$$

$$x_1 = \frac{9}{5}s - \frac{14}{5}t$$
$$x_2 = -\frac{4}{5}s - \frac{1}{5}t$$
$$x_3 = s$$
$$x_4 = t$$
where $s, t \in \mathbb{R}$

Definition

If $X_1, X_2, \ldots, X_p \in \mathbb{R}^n$ are columns with n number of entries, and if $a_1, a_2, \ldots a_p \in \mathbb{R}$ (are scalars) then $a_1X_1 + a_2X_2 + \cdots + a_pX_p$ is a linear combination of columns $X_1, X_2, \ldots, X_p$.

Example (continued)

$$\boxed{\begin{array}{l} x_1 = \frac{9}{5}a_1 - \frac{14}{5}a_2 \\ x_2 = -\frac{4}{5}a_1 - \frac{1}{5}a_2 \\ x_3 = a_1 \\ x_4 = a_2 \\ \text{where } a_1, a_2 \in \mathbb{R} \end{array}}$$

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = a_1 \begin{bmatrix} \frac{9}{5} \\ -4/5 \\ 1 \\ 0 \end{bmatrix} + a_2 \begin{bmatrix} -14/5 \\ -1/5 \\ 0 \\ 1 \end{bmatrix} = a_1\vec{V}_1 + a_2\vec{V}_2$$

$\vec{V}_1 = \begin{bmatrix} \frac{9}{5} \\ -4/5 \\ 1 \\ 0 \end{bmatrix}$ and $\vec{V}_2 = \begin{bmatrix} -14/5 \\ -1/5 \\ 0 \\ 1 \end{bmatrix}$ are the basic solutions

and the general solution $\vec{V}_h = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix}$ is a linear combination of the basic ones.

Span of a Set of Vectors

Definition

Let $\{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_k\}$ be a set of vectors in $\mathbb{R}^n$. Then the collection of all linear combinations of these vectors is called the span of these vectors, written $\mathrm{Span}\{\vec{u}_1, \vec{u}_2, \ldots, \vec{u}_k\}$.

Problem

Describe the span of the vectors $\vec{u} = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$ *and* $\vec{v} = \begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$.

Note that any linear combination of $\vec{u}$ and $\vec{v}$ is of the form $a_1\vec{u} + a_2\vec{v} = a_1\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + a_2\begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ a_1+2a_2 \\ 2a_1+3a_2 \end{bmatrix}$ which can be thought of as a vector in the yz-plane.

Also, consider an arbitrary vector $\begin{bmatrix} 0 \\ y \\ z \end{bmatrix}$ in the yz-plane. We can write any such vector as a linear combination of $\vec{u}$ an $\vec{v}$:

$$\begin{bmatrix} 0 \\ y \\ z \end{bmatrix} = x_1\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + x_2\begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix} = (-3y+2z)\begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + (2y-z)\begin{bmatrix} 0 \\ 2 \\ 3 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 1 & 2 & y \\ 2 & 3 & z \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 2 & y \\ 0 & -1 & z-2y \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 0 & -3y+2z \\ 0 & 1 & 2y-z \end{array}\right] \rightarrow \begin{cases} x_1 = -3y+2z \\ x_2 = 2y-z \end{cases}$$

$\boxed{\text{The span}\{\vec{u}, \vec{v}\} \text{ is the yz-plane.}}$